

Rank-1 separation route for LDPE/EVOH/PET

Sequential solvent-targeted recovery at 100 C; minimum step selectivity = 11.9%

Routing sequence

Target one polymer, carry the rest forward.

LDPE + EVOH + PET

mixed multilayer stream

1

LDPE separated

carry: EVOH + PET

2

EVOH separated

carry: PET

PET recovered

all polymers assigned

Step evidence

Solvent choice and selectivity are tied to the route step.

1

Target LDPE

p-Xylene

11.9% @100 C

best observed: 89.6% @150 C

next stream: EVOH + PET

2

Target EVOH

DMSO

18.4% @100 C

best observed: 44.9% @160 C

next stream: PET

3

Assign PET residue

no solvent step

complete

ranked route: LDPE -> EVOH -> PET

Recommended order

LDPE -> EVOH -> PET

min sel. 11.9%

Solvent labels: p-Xylene = 1,4-dimethylbenzene; DMSO = dimethylsulfoxide. Values shown are operating-step selectivities at 100 C.